

WORKFLOW

STEP-1 : UCSC Genome Browser

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We support **various kinds of searching** from the tracks display address bar. You can use **HGVs terms** (NM_198056.2:c.1A>C), **gnomad style variants** (1-55051215-G-GA), **BLAT sequences** directly from the search box, **documentation**, and more.

Tools

- Genome Browser** - Interactively visualize genomic data
- BLAT** - Rapidly align sequences to the genome
- In-Silico PCR** - Rapidly align PCR primer pairs to the genome
- Table Browser** - Download and filter data from the Genome Browser
- LiftOver** - Convert genome coordinates between assemblies
- REST API** - Returns data requested in JSON format

News

- Jul. 22, 2026 - [ENCODE4 cCREs and Regulation tracks for hg38 and mm10](#)
- Jul. 16, 2026 - [Long-read Structural Variants tracks for human](#)
- Jul. 7, 2026 - [UCSC Genome Browser reaches version 500](#)
- Jul. 1, 2026 - [SNV Frequencies from 30+ cohorts on hg38](#)
- Jun. 25, 2026 - [Varaico Variants tracks updated to release 3](#)
- Jun. 23, 2026 - [Non-canonical ORFs track collection on hg38](#)

STEP-2: Selection of hg38 genome and rs2476601

genome.ucsc.edu/cgi-bin/hgSearch?search=rs2476601&db=hg38

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Search across the Genome Browser site

Step 1: Select a Genome

Human - Dec. 2013 (GRCh38) Change Genome

Currently selected: Human - Dec. 2013 (GRCh38/hg38)

Step 2: Search for a Term

rs2476601 Search

Examples

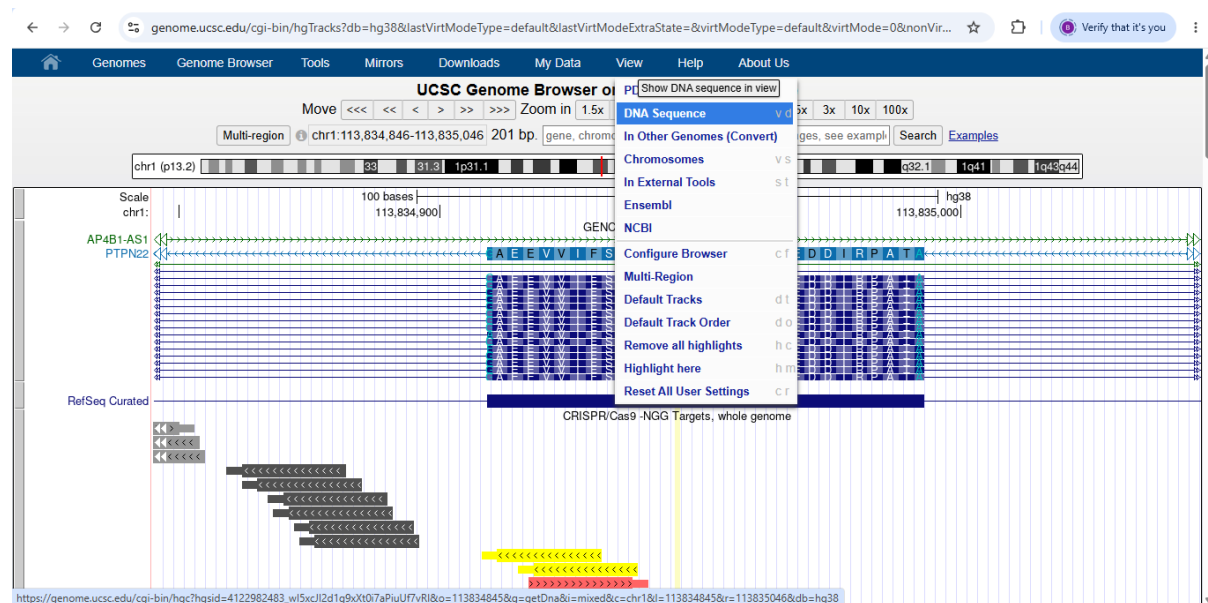
Use the tree to hide/show results from only these categories.
Hover your mouse over each category for an explanation:

- hg38 Track Data (186 results)
 - Visible Tracks (17 results)
 - All dbSNP(155) (1 results)
 - Common dbSNP(155) (1 results)
 - Common dbSNP(153) (1 results)
 - All SNPs(141) (1 results)
 - Common SNPs(141) (1 results)
 - All SNPs(142) (1 results)
 - Common SNPs(142) (1 results)
 - All SNPs(144) (1 results)
 - Common SNPs(144) (1 results)

Search Results on hg38 (Human Dec. 2013 (GRCh38/hg38))

- Common (1000 Genomes Phase 3 MAF >= 1%) Short Genetic Variants from dbSNP Release 153:
 - rs2476601 chr1:113834846-113835046
- All Short Genetic Variants from dbSNP Release 155:
 - rs2476601 chr1:113834846-113835046
- All Short Genetic Variants from dbSNP Release 153:
 - rs2476601 chr1:113834846-113835046
- NHGRI-EBI Catalog of Published Genome-Wide Association Studies:
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
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 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - rs2476601 chr1:113834696-113835196
 - Show 153 more matches for NHGRI-EBI Catalog of Published Genome-Wide Association Studies
- SNPedia all SNPs (including empty pages):
 - rs2476601 chr1:113834896-113834996
- SNPedia pages with manually typed text:
 - rs2476601 chr1:113834896-113834996
- Common (1000 Genomes Phase 3 MAF >= 1%) Short Genetic Variants from dbSNP Release 155:
 - rs2476601 chr1:113834846-113835046
- Short Genetic Variants from dbSNP Release 155 Included in ClinVar:
 - rs2476601 chr1:113834846-113835046

STEP-3: Retrieve DNA Sequence



→↻🔍 genome.ucsc.edu/cgi-bin/hg/cgi?hsid=4122982483_wl5xclJ2d1g9xXt0i7aPiUf7rVRI&o=getDna&gi=mixed&c=chr1&i=113834845&r...

☆🔖🔗🔒🔍 Verify that it's you

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Get DNA in Window (hg38/Human)

Get DNA for

Position

Note: This page retrieves genomic DNA for a single region. If you would prefer to get DNA for many items in a particular track, or get DNA with formatting options based on gene structure (introns, exons, UTRs, etc.), try using the [Table Browser](#) with the "sequence" output format. You can also use the [REST API](#) with the `/getData/sequence` endpoint function to extract sequence data with coordinates.

Sequence Retrieval Region Options:

Add extra bases upstream (5') and extra downstream (3')

Note: If a feature is close to the beginning or end of a chromosome and upstream/downstream bases are added, they may be truncated in order to avoid extending past the edge of the chromosome.

Sequence Formatting Options:

☒ All upper case.
☐ All lower case.
☐ Mask repeats: ☒ to lower case ☐ to N
☐ Reverse complement (get '-' strand sequence)

Note: The "Mask repeats" option applies only to "Get DNA", not to "Extended case/color options".

STEP-4: FASTA sequence

```
<  →  ↻  genome.ucsc.edu/cgi-bin/hg?hgsid=4122982483_wl5xclJ2d1g9xXt0i7aPiuUf7vRI&g=htcGetDna2&table=&i=mixed&o=113834845&l=11383484...  ☆  📄  🔍 Verify that it's you  ⋮

>hg38_dna range=chr1:113834846-113835046 5'pad=0 3'pad=0 strand=+ repeatMasking=none
ATTAAAGGCATGAGCCACCATGCCATCCACACTTTATTTTATACCTAC
TGAAGTGTACTCACCAGCTTCTCTCAACCAATAAATGATTCAGGTGTCC
ATACAGGAAGTGGAGGGGGGATTTCATCATCTATCCTTGGAGCAGTTGCT
ATCCAAATGTCAAAATATTGTAACAATTGTTAATTAGAACAAATCCAAA
G
```

STEP-5: Go to CRISPOR website, paste the sequence, select hg38 genome & select 20bp NGG Cas9

< → ↻ crispor.gi.ucsc.edu ☆ 📄 🔍 Verify that it's you ⋮

UC SANTA CRUZ 

CRISPOR ([citation](#)) is a program that helps design, evaluate and clone guide sequences for the CRISPR/Cas9 system. [CRISPOR Manual](#)
Mar 2026: Hopefully better reliability and speed. Many new features will be coming soon. [Full list of changes](#)

Step 1

Planning a lentiviral gene knockout screen? Use [CRISPOR Batch](#)

Sequence name (optional):

Enter a single genomic sequence, < 2300 bp, typically an exon 

[Clear Box](#) - [Reset to default](#)

```
ATTAAAGGCATGAGCCACCATGCCATCCACACTTTATTTTATACCTAC
TGAAGTGTACTCACCAGCTTCTCTCAACCAATAAATGATTCAGGTGTCC
ATACAGGAAGTGGAGGGGGGATTTCATCATCTATCCTTGGAGCAGTTGCT
ATCCAAATGTCAAAATATTGTAACAATTGTTAATTAGAACAAATCCAAA
G
```

Text case is preserved, e.g. you can mark ATGs with lowercase.
Instead of a sequence, you can paste a chromosome range, e.g. chr1:11,130,540-11,130,751

Step 2

Select a genome

Homo sapiens - Human - UCSC Dec. 2013 (GRCh38/hg38) + SNPs: dbSNP148, Kaviar

Note: pre-calculated exonic guides for this species are on the [UCSC Genome Browser](#).
We have 1527 genomes, but not yours? Search [NCBI assembly](#) and send a GCF_GCA_ID to [CRISPOR support](#).

Step 3

Select a Protospacer Adjacent Motif (PAM)

20bp-NGG - Sp Cas9, SpCas9-HF1, eSpCas9 1.1

See [notes on enzymes](#) in the manual.

SUBMIT

Version 5.2 - [Documentation](#) - [Contact us](#) - [Downloads/local installation](#) - [Citation](#) - [License](#)

STEP-6: Result Analysis.

Homo sapiens (hg38), chr1:113834846-113835046, forward genomic strand

Your input sequence is 201 bp long. It contains 21 possible guide sequences.

Shown below are their PAM sites and the expected cleavage position located -3bp 5' of the PAM site.

Click on a match for the PAM NGG below to show its 20 bp-long guide sequence. (Need help? Look at the [CRISPOR manual](#))

Colors **green**, **yellow** and **red** indicate high, medium and low specificity of the PAM's guide sequence in the genome.

Gene Models: Do not show

Variants database: dbSNP 148 Common SNPs Missing a variant database? We can add it.

Download for: [SerialCloner \(free\)](#) - [ApE \(free\)](#) - [GenomeCompiler](#) - [Benchling](#) - [SnapGene](#) - [Geneious](#) - [Vector NTI](#) - [LaserGene](#) - [Genbank](#) - [FASTA](#)

Predicted guide sequences for PAMs

Ranked by default from highest to lowest specificity score ([Hsu et al., Nat Biot 2013](#)). Click on a column title to rank by a score.

If you use this website, please cite our [paper in NAR 2018](#). Too much information? Look at the [CRISPOR manual](#).

Download as Excel tables: [Guides](#) / [Guides, all scores](#) / [Off-targets](#) / [Saturating mutagenesis assistant](#)

Position/ Strand	Guide Sequence + PAM + Restriction Enzymes + Variants	Mit Specificity Score	CFD Spec. score	Predicted Efficiency	Off-targets for 0-1-2-3-4 mismatches	Genome Browser links to matches sorted by CFD off-target score
						☐ exons only ☐ chr1 only

STEP-7: List of guide Rna's with scores.

Download as Excel tables: [Guides](#) / [Guides, all scores](#) / [Off-targets](#) / [Saturating mutagenesis assistant](#)

Position/ Strand	Guide Sequence + PAM + Restriction Enzymes + Variants ☐ Only G- ☐ Only GG- ☐ Only A-	MIT Specificity Score	CFD Spec. score	Predicted Efficiency Show all scores			Outcome		Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score ☐ exons only ☐ chr1 only
				Doench '16	Mor-Mateos	Doench-RuleSet3	Out-of-Frame	Undel		
105 / fw	AATGATTCAAGTGTCCATAC AGGB..... Cloning / PCR primers	79	89	46	52	-11	54	77	0-0-2-14-96 0-0-1-1-0 112 off-targets	4:intergenic:FBXL4-MIPS48A1 3:intergenic:KLIF4-RNU6-996P 4:intergenic:SNORA25-RP11-458K10.1 show all...
135 / rev	TTTGSATGCAACTGCTCCA AGGB..... Enzymes: <i>EthI</i> , <i>BseDI</i> , <i>LmnI</i> Cloning / PCR primers	76	89	61	41	7	54	80	0-0-2-17-106 0-0-0-0-3 125 off-targets	3:exon:PDCC2LRP11-618P17.4 4:intergenic:C2CD4B-RP11-299H22.1 4:intergenic:RALA-AC004837.1 show all...
77 / rev	GACACCTGAATCATTATTG TGGB..... Cloning / PCR primers	73	87	55	22	-35	52	74	0-0-1-12-120 0-0-1-1-3 133 off-targets	4:intron:RBMS1 3:intron:SCNN1G 4:intergenic:SLC2A2-TN1K show all...
138 / fw	GGGATTTTCATCATCTCCTG TGGB..... ⚠ Inefficient Enzymes: <i>EthI</i> , <i>BseDI</i> Cloning / PCR primers	72	91	47	23	-35	54	78	0-0-1-33-438 0-0-0-0-2 472 off-targets	4:intergenic:EGFR-LANCL2 4:intergenic:RP11-115C10.1-RP11-688I9.4 3:intron:PTPN2 show all...